Milestone 3 Report: Model Optimization & Refinement

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1. Introduction

Milestone 3 represents a critical phase in the machine learning pipeline where the focus shifts from simply building models to **optimizing, refining, and understanding their performance**. In previous milestones, baseline models for both **object detection (YOLOv8)** and **image classification (ResNet18)** were developed. However, these baseline models often do not provide optimal results due to default configurations and limited tuning.

The primary goal of this milestone is to enhance model performance through systematic experimentation and analysis. This involves adjusting key parameters, improving data quality through augmentation, leveraging pretrained knowledge using transfer learning, and evaluating the models both quantitatively and qualitatively.

Additionally, this milestone emphasizes **model interpretability**, which is especially important in medical imaging tasks such as chest X-ray analysis. Techniques like **Grad-CAM** help in understanding how the model makes decisions, thereby increasing trust and reliability.

Overall, this milestone bridges the gap between a working model and a **well-optimized, explainable, and production-ready system**.

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2. Objectives

The objectives of this milestone are designed to systematically improve and validate the model's performance. The first objective is to perform **hyperparameter tuning**, which involves adjusting parameters such as learning rate, batch size, and number of epochs to find the optimal configuration for training.

The second objective is to implement **advanced data augmentation techniques using Albumentations**. This helps increase dataset diversity and improves the model's ability to generalize to unseen data.

Another important objective is to apply **transfer learning**, where pretrained models are fine-tuned for the specific medical dataset. This reduces training time and improves accuracy, especially when the dataset is limited.

The milestone also requires conducting a **comprehensive evaluation** using validation datasets. This ensures that the model is not only learning but also generalizing effectively.

Finally, the milestone aims to provide a **detailed analysis of model strengths and weaknesses**, helping in identifying areas for improvement.

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3. Dataset Description

The dataset used in this project consists of **chest X-ray images categorized into 14 disease classes**. These classes include conditions such as Aortic Enlargement, Atelectasis, Cardiomegaly, Pleural Effusion, Pneumothorax, and others.

The dataset is structured into three main subsets:

- **Training Set**: Used to train the model
- **Validation Set**: Used to evaluate performance during training
- **Test Set**: Used for final evaluation

Each image is labeled according to the disease category, and in the case of detection models, bounding boxes are provided to indicate regions of interest.

One of the challenges with this dataset is **class imbalance**, where some diseases have significantly fewer samples than others. This can affect model performance and requires careful handling through augmentation and tuning.

The dataset is preprocessed by resizing images, normalizing pixel values, and organizing them into appropriate directory structures for training.

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4. Methodology

4.1 Model Selection

Two types of models were used in this milestone:

- **YOLOv8 (Detection Model)**: Used for detecting and localizing diseases in X-ray images using bounding boxes.
- **ResNet18 (Classification Model)**: Used for classifying images into one of the 14 disease categories.

YOLOv8 was chosen due to its efficiency and real-time detection capabilities, while ResNet18 was selected for its simplicity and effectiveness in classification tasks.

4.2 Transfer Learning

Transfer learning plays a crucial role in this project. Instead of training models from scratch, pretrained weights (trained on large datasets like ImageNet) were used.

In the classification model:

- The final fully connected layer was replaced to match 14 output classes.
- The model was fine-tuned on the chest X-ray dataset.

Benefits of transfer learning include:

- Faster convergence
- Better performance with limited data
- Reduced computational cost

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5. Hyperparameter Tuning

5.1 Detection Model (YOLOv8)

Multiple experiments were conducted to improve detection performance:

Experiment	Epochs	Batch Size	Learning Rate	mAP@50
Baseline	15	16	0.01	0.23058
Batch Tuning	20	32	0.01	0.24346
LR Tuning	20	16	0.005	0.25121
Augmentation	20	16	0.005	0.20984

Analysis:

- Increasing batch size improved stability and performance.
- Reducing learning rate led to better convergence.
- Augmentation reduced performance slightly, indicating need for better tuning.

5.2 Classification Model

For classification:

- Learning rate was adjusted
- Batch size was optimized
- Augmentation was applied

These changes resulted in improved accuracy and reduced overfitting.

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6. Data Augmentation (Albumentations)

Albumentations was used to apply advanced augmentation techniques such as:

- Horizontal flipping
- Brightness and contrast adjustment
- Rotation
- Scaling

These transformations simulate real-world variations and improve model robustness.

The impact of augmentation includes:

- Increased dataset diversity
- Reduced overfitting
- Better generalization

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7. Model Evaluation

Evaluation was performed using validation datasets.

Detection Metrics

- mAP@50
- Precision
- Recall

Classification Metrics

- Accuracy
- Loss
- Confusion Matrix

The tuned models showed improved performance compared to baseline models.

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8. Grad-CAM Visualization

Grad-CAM was implemented for the classification model to visualize important regions in the image.

This technique generates heatmaps that highlight areas influencing the model's prediction.

Observations:

- The model focuses on lung regions
- Helps validate correctness of predictions
- Improves interpretability

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9. Bounding Box Visualization

For the detection model:

- Bounding boxes were visualized on images
- Predictions were compared with ground truth

This helped in:

- Identifying false positives
- Verifying detection accuracy
- Improving qualitative analysis

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10. Results Summary

- Detection mAP improved from **0.23 to 0.25**
- Classification accuracy improved after tuning
- Grad-CAM provided interpretability
- Visualization improved model understanding

11. Strengths of the Model

- Good performance on common disease classes
- Effective use of transfer learning
- Improved generalization after tuning
- Interpretability achieved using Grad-CAM

12. Weaknesses of the Model

- Poor performance on rare classes
- Augmentation needs better tuning
- Some incorrect detections
- Slight overfitting observed

13. Conclusion

Milestone 3 successfully enhanced model performance through systematic optimization techniques. Both detection and classification models showed measurable improvements.

The use of transfer learning, hyperparameter tuning, and augmentation significantly contributed to better performance. Additionally, interpretability techniques like Grad-CAM added value by making model decisions understandable.

14. Final Outcome

- Optimized detection and classification models
- Improved performance metrics
- Enhanced interpretability
- Successful completion of Milestone 3 objectives